INCH-POUND
MIL-DTL-23053/17B
17 June 1996
SUPERSEDING
MIL-I-23053/17A
01 May 1986

DETAIL SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, FLAME RETARDED, MODIFIED POLYOLEFIN, FLEXIBLE, CROSSLINKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the sleeving described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-DTL-23053.

REQUIREMENTS:

Polymer type: The base polymer used in formulating this sleeving shall be a polyolefin.

Continuous operating temperature range: -55°C (-67°F) to +120°C (+250°F).

Classification: The heat shrinkable sleeving shall be furnished in the following classes, as specified (see 6.2a):

Class 1 - Standard wall
Class 2 - Thin wall

Color: Translucent to clear.

Longitudinal Change: Classes 1 and 2: +2, -10 percent

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:

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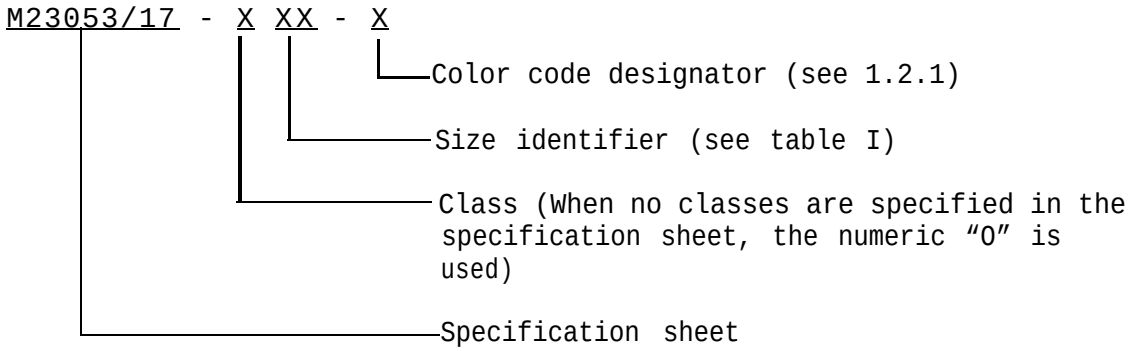

Example: Class 1, 0.250 inch (6.35 mm) as supplied ID sleeving shall be identified as M23053/17-106-C.

TABLE I. Construction details, inches (mm). 1/

Military part number	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
Class 1			
M23053/17-101-C	.046(1.17)	.023(.58)	.016 ± .003(.406 ± .076)
M23053/17-102-C	.063(1.60)	.031(.79)	.017 ± .003(.432 ± .076)
M23053/17-103-C	.093(2.36)	.046(1.17)	.020 ± .003(.508 ± .076)
M23053/17-104-C	.125(3.18)	.062(1.58)	.020 ± .003(.508 ± .076)
M23053/17-105-C	.187(4.75)	.093(2.36)	.020 ± .003(.508 ± .076)
M23053/17-106-C	.250(6.35)	.125(3.18)	.025 ± .003(.635 ± .076)
M23053/17-107-C	.375(9.53)	.187(4.75)	.025 ± .003(.635 ± .076)
M23053/17-108-C	.500(12.7)	.250(6.35)	.025 ±.003(.635 ± .076)
M23053/17-109-C	.750(19.1)	.375(9.53)	.030 ± .003(.762 ± .076)
M23053/17-110-C	1.000(25.4)	.500(12.7)	.035 ± .005(.889 ± .127)
Class 2			
M23053/17-201-C	.046(1.17)	.023(.58)	.010 ± .003(.254 ± .076)
M23053/17-202-C	.063(1.60)	.031(.79)	.010 ± .003(.254 ± .076)
M23053/17-203-C	.093(2.36)	.046(1.17)	.012 ± .003(.305 ± .076)
M23053/17-204-C	.125(3.18)	.062(1.58)	.012 ± .003(.305 ± .076)
M23053/17-205-C	.187(4.75)	.093(2.36)	.012 ± .003(.305 ± .076)
M23053/17-206-C	.250(6.35)	.125(3.18)	.014 ± .003(.356 ± .076)
M23053/17-207-C	.375(9.53)	.187(4.75)	.014 ± .003(.356 ± .076)
M23053/17-208-C	.500(12.7)	.250(6.35)	.014 ±.003(.356 ± .076)
M23053/17-209-C	.750(19.1)	.375(9.53)	.017 ± .003(.432 ± .076)
M23053/17-210-C	1.000(25.4)	.500(12.7)	.020 ± .005(.508 ± .127)

1/ Diameter limits for object to be enclosed shall be as recommended in technical data.

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2/ Wall thickness values are less when shrinkage is restricted.

Unrestricted shrinkage: Test method 4.6.5: 200° ± 2°C (392° ± 4°F) for 3 minutes.

TABLE II. Physical properties. 1/

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table 1	4.6.3
Heat shock	No cracks, flowing or dripping	4.6.8 200° ± 3°C (392° ± 6°F)
Secant modulus, psi (MPa), minimum	15,000(103.4)	4.6.12.1, 2 percent strain, ASTM D882
Restricted shrinkage	No cracks	4.6.6.1.1 200° ± 2°C (392° ± 4°F)
Voltage withstand	Pass	4.6.6.2
Clarity stability	Pass	4.6.16 125° ± 1°C (257° ± 2°F), 24 hours
Concentricity, min.	70%	4.6.3.3
After unrestricted shrinkage:		
ID, maximum	Table 1	4.6.3
Wall thickness	Table 1	4.6.3
Tensile strength, psi (MPa), minimum	1,500(10.4)	4.6.13 ASTM D638, 20 inches/rein.
Ultimate elongation, percent, minimum	200	4.6.13 ASTM D638, 20 inches/rein.
Dielectric strength, volts/roil (Kv/mm), minimum	750(29.5)	4.6.2 ASTM D2671

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test procedure and conditions
Volume resistivity, ohm-cm, minimum	10^{14}	4.6.2 ASTM D876
Dielectric constant, maximum	3.7	4.6.2 ASTM D150
Specific gravity, maximum	1.35	4.6.2 ASTM D792
Water absorption, percent, maximum	0.5	4.6.2 ASTM D570, 24 hrs. at 23°C
Corrosion	No corrosion	4.6.10.1 and 4.6.10.2 120° ± 2°C (248° ± 4°F), 16 hours
Low temperature flexibility	No cracking	4.6.7.1 -55° ± 1°C (-67° ± 2°F)
Flammability	Self-extinguishing within 15 seconds; 3 inches	4.6.14 Procedure A ASTM D2671
Heat resistance, property after:		4.6.9 150° ± 2±C (302° ± 4°F), 168 hours
Ultimate elongation, percent, minimum	100	
Fluid resistance, property after:		4.6.11
Tensile strength, psi (MPa), minimum	1000(6.9)	
Elongation, percent, minimum	100	
Dielectric strength, volt/roil (Kv/mm), minimum	500(12.7)	
Fungus resistance	Non-nutrient	4.6.2 ASTM G21

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1/ Unless otherwise specified, requirements, test procedures, and conditions are for all classes.

Storage life conditions: Supplier shall certify to storage at 18° to 35°C (65° to 95°F) for 2 years. Conformance to 3.5. See 3.5.2 for storage life extension.

Intended use:

Class 1 sleevings are intended for normal use where flame retarded translucent to clear materials are required.

Class 2 is intended where thin wall product still provides for adequate electrical insulation and confined spaces require smaller wire diameters.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Preparing activity:
Navy - AS

(Project 5970-1129-17)

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